IOT Based Seashore Waste Collector Boat

A MINI PROJECT II REPORT

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ABSTRACT

Water pollution caused by floating plastic waste and debris has become a serious environmental problem in rivers, lakes, and coastal regions. Manual cleaning of water bodies is often difficult, time-consuming, and requires significant human effort. To address this issue, this project proposes the design and development of an **IoT-Based Seashore Waste Collector Boat** that can efficiently collect floating waste from water surfaces.

The system is designed using an **ESP32 microcontroller** as the main control unit, which enables wireless communication and control through the **Blynk IoT mobile application**. The boat moves on the water using **DC gear motors** controlled by a motor driver module. A **floating net mechanism** is attached to the boat to collect waste such as plastic bottles, wrappers, and other floating debris as the boat moves forward. To assist the operator, an **ESP32-CAM module** is integrated to provide live video streaming, allowing real-time monitoring of the cleaning process. A **servo motor** is used to rotate the camera so that both the front navigation view and the waste collection net can be observed.

The system is powered using a **12V battery supported by a solar charging system**, which makes the project energy-efficient and environmentally friendly. By combining robotics, IoT communication, and renewable energy, the proposed system provides a cost-effective and practical solution for improving water surface cleanliness and reducing environmental pollution.

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW OF WATER POLLUTION IN COASTAL AREAS

Coastal regions and seashores are increasingly affected by pollution caused by plastic waste, bottles, food wrappers, and other floating debris. These wastes enter water bodies through human activities, tourism, drainage systems, and improper waste disposal. Over time, this accumulation of waste affects marine ecosystems, harms aquatic life, and reduces the environmental quality of coastal regions.

Manual cleaning of water surfaces is difficult and time-consuming. Workers often have to use boats or nets to collect floating waste, which requires continuous human effort and may not be efficient for large areas. Because of this, there is a growing need for automated systems that can assist in collecting floating waste from water bodies.

Technological solutions such as robotic systems and IoT-based devices can help address this issue. Automated waste collection systems can improve cleaning efficiency while reducing human effort and operational costs.

1.2 ROLE OF AUTOMATION IN ENVIRONMENTAL CLEANING

Automation plays a significant role in solving modern environmental challenges. Robotics and embedded systems are increasingly being used in applications such as waste management, environmental monitoring, and water cleaning. These systems are designed to perform tasks automatically with minimal human intervention.

Automated cleaning systems can continuously operate on water surfaces and collect floating waste effectively. By integrating sensors, motors, and wireless communication technologies, such systems can detect waste, collect it, and transport it to a storage container.

The use of microcontrollers such as ESP32 allows these systems to be controlled remotely through mobile applications. This makes it easier for users to operate and monitor the system from a safe distance. Automation also helps improve accuracy, efficiency, and reliability in waste collection processes.

1.3 IMPORTANCE OF SMART WASTE COLLECTION SYSTEMS

Traditional waste collection methods in water bodies rely heavily on manual labor and mechanical tools. These methods are often inefficient and cannot handle large volumes of floating waste. In addition, manual operations may expose workers to unsafe or polluted water conditions.

Smart waste collection systems provide an effective solution to this problem. These systems use electronic components, sensors, and automated mechanisms to detect and collect waste. With the help of wireless communication technologies, operators can control and monitor the system remotely.

The integration of renewable energy sources such as solar panels further enhances the sustainability of these systems. Solar power enables the system to operate continuously without relying entirely on external power sources.

1.4 INTRODUCTION TO THE PROPOSED SYSTEM

The proposed system is an **IoT-based Seashore Waste Collector Boat** designed to remove floating waste from water surfaces. The boat is equipped with a waste collection mechanism that gathers floating materials such as plastic bottles, wrappers, and small debris.

The system uses an ESP32 microcontroller as the main control unit. The ESP32 communicates with the Blynk IoT mobile application, allowing users to control the boat wirelessly using a smartphone. The boat movement is controlled through DC motors connected to a motor driver module, while a servo motor operates the waste collection mechanism.

An ESP32-CAM module is also integrated into the system to provide live video monitoring. This allows the operator to view the surrounding environment in real time while controlling the boat. Additionally, the system uses a solar panel and battery to provide power, making the system energy efficient and environmentally friendly.

1.5 OBJECTIVES OF THE PROJECT

The main objectives of this project are:

- To design and develop an IoT-based robotic boat capable of collecting floating waste from water surfaces.
- To implement wireless control of the boat using the ESP32 microcontroller and Blynk IoT mobile application.
- To integrate a camera module for real-time monitoring of the cleaning process.
- To develop an efficient waste collection mechanism that gathers floating debris.
- To utilize solar energy for powering the system and improving sustainability.
- To demonstrate a cost-effective solution for cleaning seashores and water bodies.

CHAPTER-2

LITERATURE REVIEW

2.1 EVOLUTION OF WATER CLEANING TECHNOLOGIES AND SMART ENVIRONMENTAL SYSTEMS

The increasing level of pollution in water bodies such as rivers, lakes, and coastal areas has created serious environmental challenges worldwide. Traditionally, floating waste was removed manually using boats, nets, and other simple tools. These methods required significant human effort and were often slow and inefficient for covering large areas. As the use of plastic materials increased over time, the amount of floating waste entering water bodies also grew, making manual cleaning methods insufficient for maintaining a clean aquatic environment.

With the advancement of technology, mechanical and robotic cleaning systems were gradually introduced to improve waste collection processes. These systems used electric motors, conveyors, and other mechanisms to collect floating debris from the water surface and store it in containers. The use of microcontrollers such as Arduino and ESP32 further improved these systems by enabling better control of motors and integration with sensors. Sensors and cameras allowed the systems to detect obstacles, monitor operations, and improve the efficiency of waste collection.

In recent years, the emergence of Internet of Things (IoT) technology has enabled the development of smarter environmental cleaning systems. IoT platforms allow devices to communicate with mobile applications, making it possible to monitor and control systems remotely. In this project, these technologies are applied to develop an **IoT-based Seashore Waste Collector Boat**. The system uses an ESP32 microcontroller to control the boat's movement and integrates an ESP32-CAM for live monitoring. A floating net mechanism collects waste while the boat moves across the water surface, providing an efficient and environmentally friendly solution for cleaning water bodies.

2.2 IOT BASED SURFACE CLEANING ROBOT

Experiment	Problem Gap	Result
A floating robotic system was developed to collect plastic waste and debris from water bodies using a microcontroller-based control unit. The robot used DC motors for propulsion and a mechanical waste collector to gather floating materials from the water surface. Wireless communication was used for remote control.	Traditional water cleaning methods require manual labor and are inefficient for large water bodies. Existing automated systems lack real-time monitoring and remote control features.	The robotic system successfully collected floating waste and demonstrated improved cleaning efficiency. Wireless control allowed operators to monitor and operate the robot remotely.

Author(s): R. Kumar, P. Sharma

Journal: International Journal of Environmental Robotics, 2021

Summary: This research focuses on the development of a robotic system for collecting floating waste using microcontroller-based automation.

2.2.1 Learning and Relevance to Project

Key Learning

- Robotic systems can efficiently collect floating waste from water surfaces.
- Wireless communication allows remote control of cleaning robots.
- Microcontroller-based control systems improve automation and reliability.
- Automated systems reduce human labor and increase cleaning efficiency.
- Integration of sensors helps detect obstacles and improve system performance.

Relevance to Project

- The proposed project also uses a robotic boat to remove floating waste from water bodies.
- IoT communication is used to control the boat remotely through a mobile application.
- The system uses an ESP32 microcontroller for controlling motors and operations.
- Automation reduces manual effort in cleaning polluted water surfaces.
- The design supports environmental protection by removing plastic waste from water.

2.3 SOLAR POWERED FLOATING WASTE COLLECTOR

Experiment	Problem Gap	Result
A solar-powered floating robot was designed to collect plastic waste from rivers and lakes. The robot used solar panels to generate electricity and power DC motors for movement and waste collection.	Many robotic cleaning systems rely on battery power and require frequent charging. This limits operational time and efficiency.	The solar-powered robot operated continuously under sunlight and effectively collected floating waste while reducing energy consumption.

Author(s): S. Patel, A. Shah

Journal: Renewable Energy Robotics Journal, 2020

Summary: This study demonstrates the use of solar energy to power robotic waste collection systems.

2.3.1 Learning and Relevance to Project

Key Learning

- Solar energy can provide sustainable power for robotic systems.
- Renewable energy reduces dependency on external power sources.
- Solar-powered systems can operate for longer durations in outdoor environments.
- Energy-efficient robots are suitable for environmental applications.

Relevance to Project

- The proposed waste collector boat also integrates solar energy for power supply.
- Solar energy improves sustainability of the environmental cleaning system.
- The boat can operate for longer periods using renewable energy.
- The system becomes eco-friendly by reducing electricity consumption.
- Solar panels support continuous operation in outdoor water environments.

2.4 IOT BASED SMART WASTE MANAGEMENT SYSTEM

Experiment	Problem Gap	Result
An IoT-based smart waste management system was developed using sensors and wireless communication. The system improved efficiency in waste management by reducing manual inspection and enabling better planning of cleaning operations. Such IoT-based solutions demonstrate how smart technologies can be used to improve environmental monitoring and waste control in modern urban areas.	Traditional waste management systems lack real-time monitoring and efficient waste tracking mechanisms. This often leads to delays in identifying waste accumulation and responding to cleaning needs. As a result, waste may remain in the environment for longer periods, causing pollution and affecting ecological balance.	The system enabled real-time monitoring of waste levels and improved the efficiency of waste management operations. It allowed users to track waste accumulation remotely and respond quickly when cleaning was required. This approach reduced manual inspection and helped maintain cleaner environments through timely waste collection.

Author(s): A. Singh, M. Verma

Journal: Journal of Smart Environmental Systems, 2019

Summary: This research focuses on IoT technology for smart waste monitoring and management.

2.4.1 Learning and Relevance to Project

Key Learning

- IoT technology enables real-time monitoring of waste management systems.
- Wireless communication allows remote control of embedded devices.
- Mobile applications improve user interaction with hardware systems.
- IoT platforms help in monitoring environmental conditions effectively.
- Smart waste management systems increase operational efficiency.

Relevance to Project

- The proposed system uses IoT technology for controlling the waste collector boat.
- Blynk IoT application is used to operate the boat through a smartphone.
- Wireless communication enables real-time control and monitoring.
- IoT integration improves system automation and efficiency.
- The system supports smart environmental cleaning solutions

2.5 CAMERA BASED MONITORING FOR ROBOTIC SYSTEMS

Experiment	Problem Gap	Result
A robotic monitoring system was developed using a camera module that provided live video streaming. The robot transmitted real-time video to a mobile device for monitoring its surroundings.	Many robotic systems lack visual monitoring capabilities, making navigation difficult in complex environments.	The system successfully provided real-time video streaming and improved remote monitoring of robotic operations.

Author(s): J. Thomas, R. George

Journal: Robotics Vision Systems Journal, 2022

Summary: The study focuses on integrating camera modules for monitoring robotic systems.

2.5.1 Learning and Relevance to Project

Key Learning

- Camera modules provide real-time monitoring of robotic operations.
- Live video streaming helps operators control robots more effectively.
- Visual feedback improves navigation and obstacle detection.
- Camera-based monitoring increases system safety and reliability.
- Real-time video helps identify waste locations in water bodies.

Relevance to Project

- The proposed system uses an ESP32-CAM module for live monitoring.
- The camera allows the operator to observe the surrounding environment.
- Live video helps guide the boat towards floating waste.
- Camera integration improves remote operation of the robotic system.
- Visual monitoring increases the efficiency of waste collection.

2.6 AUTONOMOUS WATER SURFACE CLEANING ROBOT

Experiment	Problem Gap	Result
An autonomous water cleaning robot was developed using sensors and microcontrollers. The robot was capable of navigating water surfaces and collecting floating debris automatically. Sensors guided the movement of the robot and helped avoid obstacles in the water. The microcontroller controlled the motors for movement and waste collection. The experiment showed that the robot could effectively remove floating waste from water surfaces.	Manual cleaning of water bodies is inefficient and time-consuming. Many automated systems lack navigation capability. This limitation reduces their efficiency in collecting floating waste from large water areas. Therefore, there is a need for an intelligent system that can move efficiently and perform cleaning operations effectively.	The autonomous robot successfully navigated water surfaces and collected floating waste with minimal human intervention. The system demonstrated efficient movement and effective waste collection during testing. This result shows that the robot can be a useful solution for maintaining cleanliness in water bodies such as ponds, lakes, and rivers.

Author(s): L. Chen, Y. Wang

Journal: International Journal of Robotics Research, 2021

Summary: The research highlights the use of autonomous robots for cleaning water surfaces.

2.6.1 Learning and Relevance to Project

Key Learning

- Autonomous robots can perform environmental cleaning tasks efficiently.
- Sensors help robots detect obstacles and navigate safely.
- Automation improves operational efficiency in waste collection systems.
- Robotic systems reduce human involvement in hazardous environments.
- Intelligent control systems enhance robotic performance.

Relevance to Project

- The proposed project also focuses on automated waste collection from water bodies.
- Sensors and microcontrollers are used for controlling robotic operations.
- The system improves cleaning efficiency in polluted water environments.
- Automation reduces the need for manual waste removal.
- The project contributes to the development of smart environmental cleaning technologies.

2.7 RESEARCH GAP AND MOTIVATION FOR THIS PROJECT

Although several studies have been conducted on robotic systems for waste collection and environmental cleaning, many existing solutions still have certain limitations. Most of the previously developed water cleaning robots mainly focus on mechanical waste collection without integrating smart technologies such as Internet of Things (IoT) and real-time monitoring systems. As a result, these systems often require human supervision and are not suitable for large-scale environmental applications.

Another limitation in many existing systems is the lack of efficient remote control and monitoring mechanisms. Some robotic cleaning devices are manually operated or rely on short-range communication methods, which restrict their operational flexibility. Without proper monitoring systems, it becomes difficult for operators to observe the working condition of the robot and identify waste locations in real time.

To address these limitations, the proposed **IoT-Based Seashore Waste Collector Boat** introduces an integrated solution that combines robotic waste collection with IoT- based remote control and real-time monitoring using an ESP32-CAM module. The system aims to improve the efficiency, flexibility, and sustainability of water cleaning operations.

The motivation behind this project is to develop a **cost-effective and environmentally friendly solution** for removing floating waste from water bodies. By integrating robotics and IoT technology, the system supports modern environmental cleaning methods and contributes to sustainable water resource management.

CHAPTER - 3

PROBLEM IDENTIFICATION & METHODOLOGY

3.1 PROBLEM IDENTIFICATION

Water pollution caused by floating waste has become a major environmental concern in many parts of the world. Plastic bottles, wrappers, thermocol pieces, and other debris often accumulate on the surface of water bodies such as rivers, lakes, ponds, and coastal areas. This waste not only affects the beauty of natural water bodies but also harms aquatic life and disrupts the ecological balance.

Traditional methods of cleaning water bodies mainly depend on manual labor, where workers use boats or nets to collect floating waste. These methods are time-consuming, inefficient, and sometimes dangerous due to polluted water conditions. In addition, manual cleaning cannot cover large areas effectively and requires continuous human effort.

Therefore, there is a need for an efficient and automated solution that can assist in collecting floating waste from water surfaces. The proposed **IoT-Based Seashore Waste Collector Boat** is designed to address this problem by providing a remotely controlled robotic system capable of collecting waste with minimal human intervention.

3.2 OBJECTIVES

The primary objective of this project is to design and develop a **smart waste collection boat** capable of removing floating waste from water surfaces using IoT technology.

The main objectives include:

- To design a floating robotic boat that can move across water bodies.
- To collect floating waste using a **net-based waste collection mechanism**.
- To implement remote control of the boat using the **Blynk IoT platform**.
- To provide real-time monitoring through the **ESP32-CAM module**.
- To use a **solar-powered charging system** to support energy-efficient operation.
- To reduce human effort and improve the efficiency of water surface cleaning.

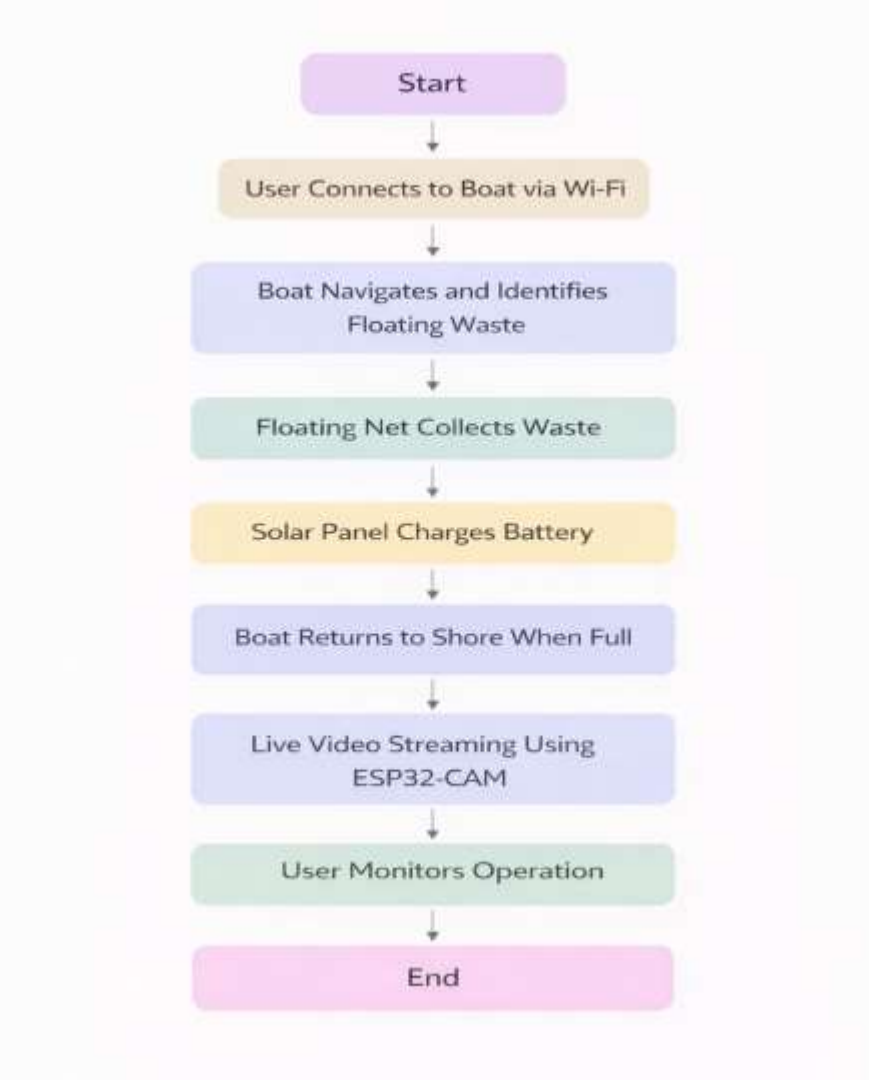
3.3 METHODOLOGY

The methodology of this project involves designing a robotic boat capable of moving across water surfaces and collecting floating waste using a net-based mechanism. The system is controlled by an ESP32 microcontroller and operated through an IoT mobile application.

The overall system consists of four main sections:

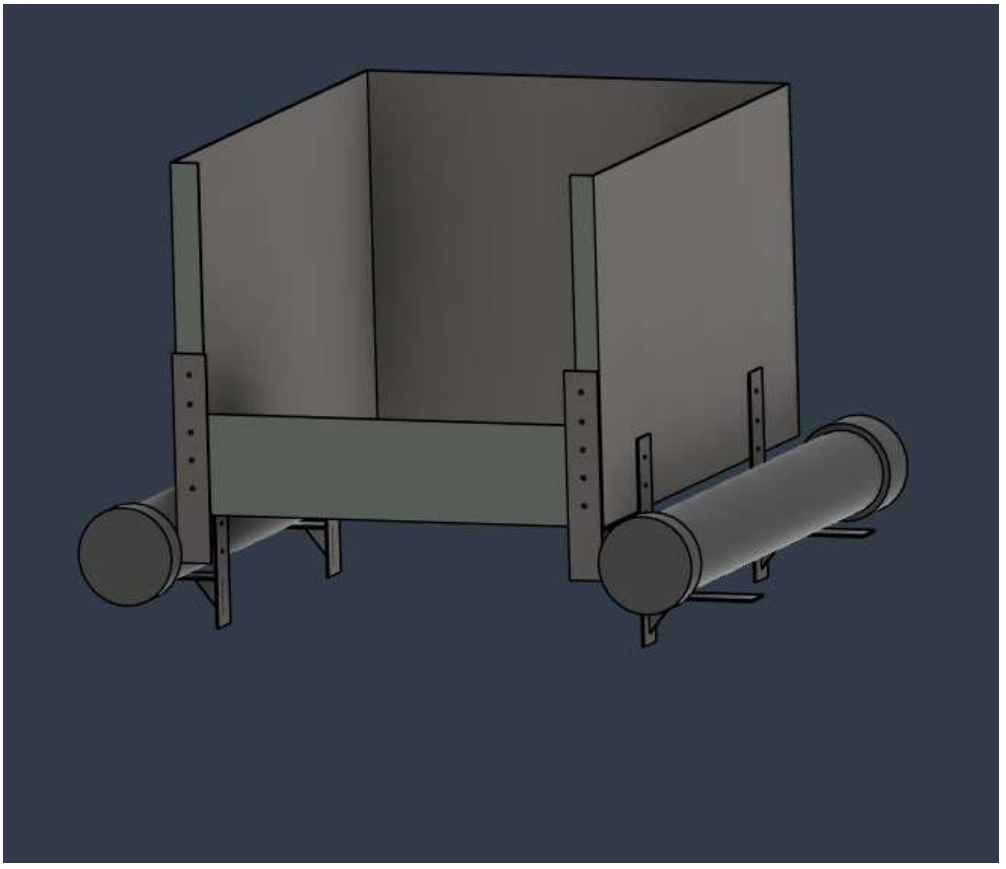
- Power supply system
- Control system
- Propulsion system
- Waste collection mechanism

The ESP32 microcontroller receives commands from the mobile application through Wi-Fi and controls the movement of the boat using a motor driver module. The DC motors drive the propellers, enabling the boat to move forward or backward. As the boat moves across the water surface, floating waste is captured using a net placed at the front of the boat. The ESP32-CAM module provides live video streaming so that the operator can monitor the cleaning process in real time.



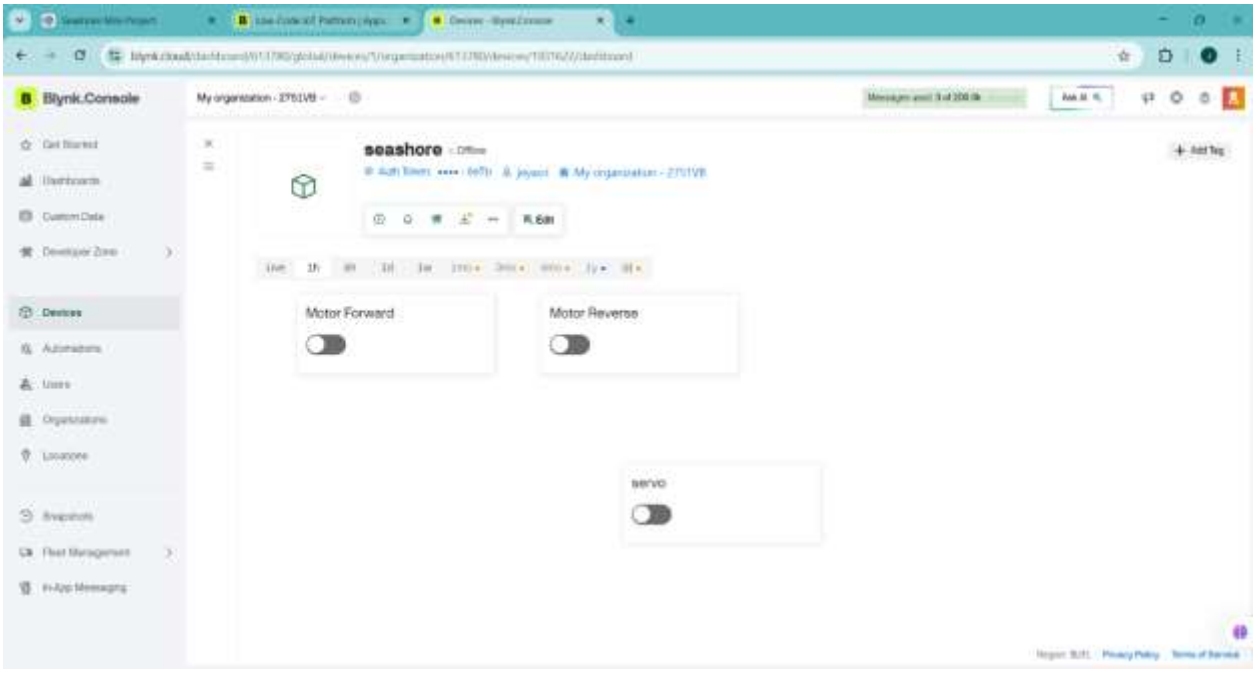
3.3.1 Hardware Setup and Boat Design

The hardware setup consists of a floating boat structure designed to carry all the required electronic components. The main components include the ESP32 microcontroller, motor driver module, DC gear motors, servo motor, ESP32-CAM module, solar panel, battery, and solar charge controller. The boat structure is designed to maintain balance and stability while moving on water. The DC motors with propellers are attached at the rear side of the boat to provide propulsion. A floating net mechanism is installed at the front of the boat to collect waste while the boat moves forward.



3.3.2 IoT Control System using ESP32

The ESP32 microcontroller acts as the main control unit of the system. It manages communication between the mobile application and the hardware components of the boat. The ESP32 connects to the Wi-Fi network and communicates with the **Blynk IoT** application, which allows the user to control the movement of the boat remotely. Through the application interface, the operator can send commands to move the boat forward or reverse. The ESP32 processes these commands and controls the motor driver module accordingly.



3.3.3 Electrical Circuit Design

The electrical circuit integrates all the electronic components required for the operation of the waste collector boat. The **L298N motor driver module** is used to control the DC motors, while the ESP32 microcontroller sends control signals to the motor driver. A **12V lead-acid battery** supplies power to the system, and a **solar charge controller** manages the charging of the battery through a solar panel. The ESP32-CAM module is connected to the ESP32 for real-time video streaming, while a servo motor is used to adjust the camera position.

3.3.4 System Integration and Operation

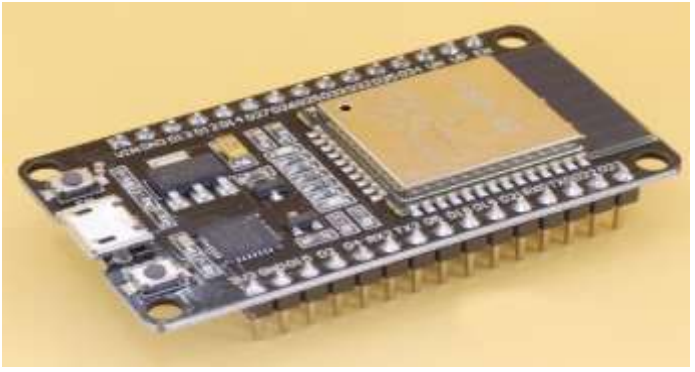
All hardware components are integrated to form a complete waste collection system. Once the system is powered on, the ESP32 connects to the Wi-Fi network and establishes communication with the Blynk IoT application. The operator uses the mobile application to control the boat's movement while observing the live video feed provided by the ESP32-CAM module. As the boat moves forward, floating waste enters the net placed at the front of the boat. When the net becomes full, the operator navigates the boat back to the shore for waste removal.

3.4 THEORETICAL BACKGROUND OF COMPUTER USED

3.4.1 ESP32 Microcontroller Overview

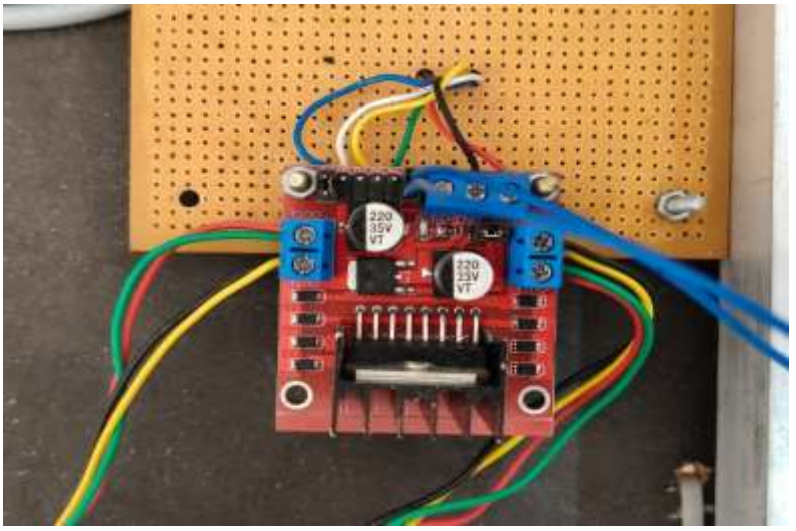
The ESP32 microcontroller is a powerful embedded system widely used in IoT applications. It provides built-in Wi-Fi and Bluetooth connectivity, making it suitable for wireless communication and remote monitoring systems.

In this project, the ESP32 serves as the main controller that manages communication between the mobile application and the hardware components. It processes user commands, controls the motor driver module, and coordinates the operation of other connected devices.



3.4.2 Motor Driver (L298N) Working Principle

The L298N motor driver module is used to control the direction and speed of DC motors. It acts as an interface between the microcontroller and the motors. The ESP32 sends digital control signals to the motor driver inputs. Based on these signals, the motor driver regulates the current supplied to the motors, enabling forward and reverse movement of the boat.



3.4.3 Solar Power System

The solar power system consists of a solar panel, solar charge controller, and battery. The solar panel converts sunlight into electrical energy, which is used to charge the battery through the charge controller. The solar charge controller regulates the charging process and prevents overcharging or deep discharge of the battery. This renewable energy system makes the project environmentally friendly and reduces dependency on external power sources.



3.4.4 DC Gear Motor and Propeller Mechanism

DC gear motors are used to provide propulsion for the boat. These motors convert electrical energy into mechanical motion. The attached propellers push water backward, generating forward movement of the boat. Gear motors are preferred because they provide higher torque at lower speeds, which is suitable for water-based movement.



3.4.5 ESP32-CAM Module Working Principle

The ESP32-CAM module is used in the system to provide real-time visual monitoring of the waste collection process. It is a compact camera module integrated with the ESP32 microcontroller, which supports Wi-Fi communication and image processing capabilities. The module captures images and streams video through a wireless network, allowing the operator to observe the surroundings of the boat remotely. In this project, the ESP32-CAM continuously captures frames from the camera sensor and transmits them as a live video stream through the Wi-Fi network. The operator can access the video feed using a web browser or mobile device by entering the IP address generated by the module. This live monitoring feature helps the user identify floating waste and monitor the filling level of the collection net, enabling better navigation and efficient operation of the waste collector boat.



3.4.6 Solar Charge Controller Working Principle

The solar charge controller is an important component used to regulate the charging process of the battery from the solar panel. It acts as a protective device that controls the flow of electrical energy from the solar panel to the battery. The main purpose of the charge controller is to prevent overcharging, over-discharging, and voltage fluctuations that could damage the battery.

In this project, the solar panel generates electrical energy from sunlight and sends it to the charge controller. The charge controller then regulates this power and safely charges the 12V lead-acid battery. It also ensures that the battery receives the correct voltage and current levels. By controlling the charging process, the charge controller improves battery life and provides a stable power supply for operating the waste collector boat.



3.4.7 12V Lead-Acid Battery Working Principle

The 12V lead-acid battery is used as the main power source for the waste collector boat. It stores electrical energy and supplies power to different electronic components such as the ESP32 microcontroller, motor driver module, DC motors, and other devices used in the system.

The battery stores energy in the form of chemical energy and converts it into electrical energy when connected to the circuit. In this project, the battery is charged using a solar panel through a solar charge controller. During operation, the stored energy is supplied to the motor driver and control circuits, enabling the boat to move and perform waste collection tasks. The use of a rechargeable lead-acid battery provides reliable power and supports continuous operation of the system.



3.4.8 Servo Motor Working Principle

The servo motor is used in the system to control the movement of the floating waste collection net. A servo motor is a rotary actuator that allows precise control of angular position. It operates based on pulse width modulation (PWM) signals generated by a microcontroller.

In this project, the ESP32 microcontroller sends PWM control signals to the servo motor to adjust its position. Based on the command received from the mobile application through the IoT platform, the servo motor rotates to a specific angle. This movement helps in opening or adjusting the position of the floating net mechanism used for collecting waste from the water surface. The servo motor provides accurate and controlled motion, which improves the efficiency of the waste collection process and ensures proper operation of the system.



3.5 SUMMARY

This chapter explained the problem identification, objectives, and methodology of the proposed IoT-based seashore waste collector boat. The design approach, hardware setup, electrical circuit design, and system integration were also discussed. In addition, the theoretical background of the major components used in the project was presented. These elements together form the foundation for developing an efficient water surface waste collection system.

In addition, the theoretical background of the major components used in the project was presented to provide a clear understanding of their functionality and importance in the system. The working principles of the control unit, propulsion mechanism, and power supply system were explained to highlight how each component contributes to the overall operation of the boat.

Furthermore, the methodology section outlined the step-by-step process involved in developing the prototype, including hardware integration and IoT-based communication. The use of real-time monitoring and remote control enhances the usability and effectiveness of the system in practical applications. Overall, this chapter provides the necessary foundation for implementing and analyzing the proposed waste collection system in the following chapters.

CHAPTER 4
IMPLEMENTATION FRAMEWORK

4.1 SYSTEM ARCHITECTURE

The proposed **IoT-Based Seashore Waste Collector Boat** is designed as an integrated system that combines hardware components, embedded control, and IoT communication. The architecture mainly consists of four major sections: the power system, control unit, propulsion mechanism, and waste collection mechanism. Each section works together to enable efficient waste collection from water surfaces.

The power system provides the required electrical energy to operate the boat using a battery and solar charging system. The control unit, based on the ESP32 microcontroller, manages the communication between the mobile application and the boat. The propulsion system includes DC gear motors and propellers that enable movement on water. The waste collection mechanism consists of a floating net that collects as the boat moves forward.

4.1.1 Hardware Components

The hardware components used in this project are selected to ensure efficient operation of the waste collector boat. The primary control unit is the **ESP32 microcontroller**, which handles wireless communication and motor control. A **motor driver module (L298N)** is used to control the direction and movement of the DC motors. The propulsion of the boat is achieved using **two DC gear motors connected to propellers**, allowing the boat to move smoothly across water surfaces. An **ESP32-CAM module** is integrated to provide real-time video monitoring of the waste collection process. A **servo motor** is used to adjust the camera angle for better visibility. The system is powered using a **12V lead-acid battery**, and a **solar charge controller with a solar panel** is used to charge the battery using renewable energy. This combination of components enables the boat to operate efficiently while maintaining low power consumption.

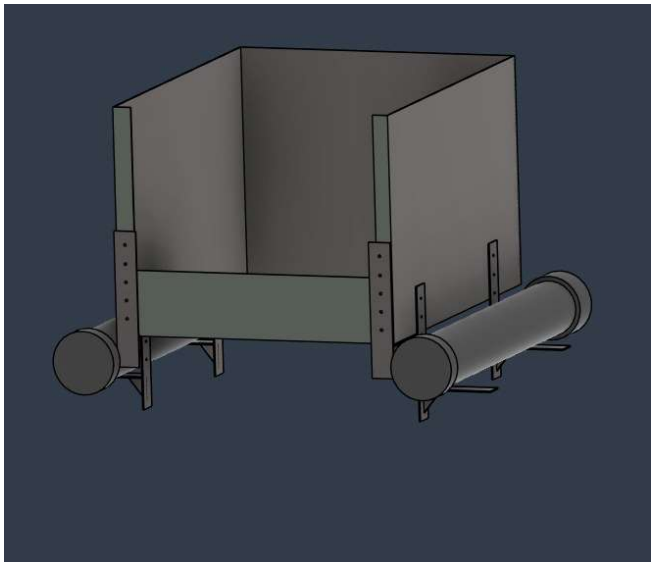
4.1.2 Software Stack

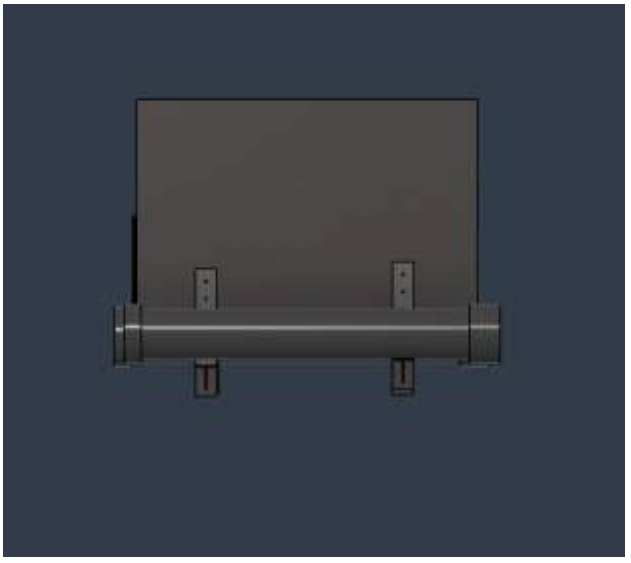
The software part of the system is responsible for controlling the hardware components and enabling communication between the boat and the mobile device. The program is developed using the **Arduino IDE**, which supports programming the ESP32 microcontroller. The **Blynk IoT platform** is used to create a mobile interface that allows the user to control the boat remotely. Through this application, commands such as forward movement, reverse movement, and camera control can be transmitted to the ESP32 microcontroller. The ESP32 receives these commands through Wi-Fi and processes them to control the motor driver module and servo motor accordingly. In addition, the ESP32-CAM module streams real-time video to the user, allowing continuous monitoring of the waste collection operation.

4.2 PROTOTYPE DESIGN

The prototype of the waste collector boat is designed to demonstrate the working concept of the system. The structure of the boat is built using lightweight materials to ensure buoyancy and stability on the water surface.

The electronic components are mounted securely on the boat platform to prevent water damage and maintain proper balance. The propulsion motors are placed at the rear side of the boat, while the waste collection net is positioned at the front to collect floating debris during forward motion. The design also ensures easy access to electronic components for maintenance and testing purposes.

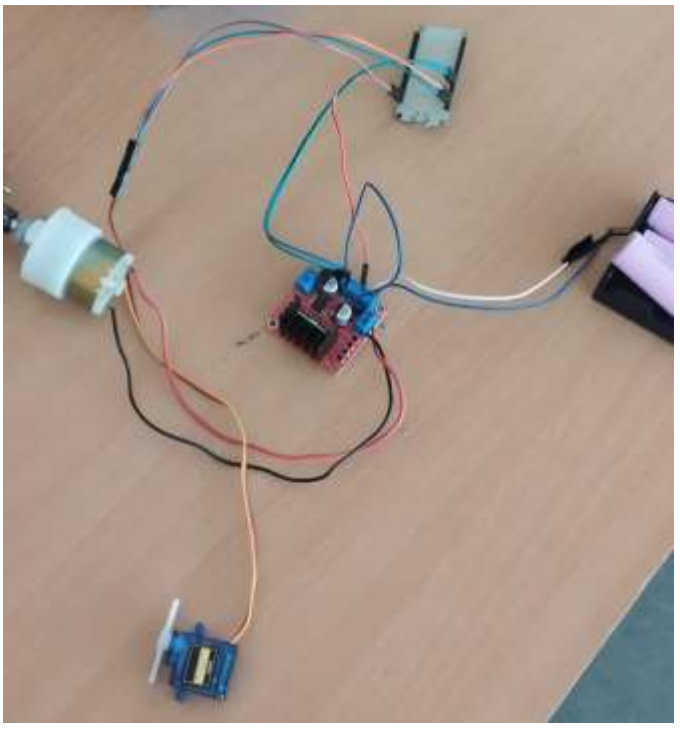




4.3 REAL-TIME EXPERIMENTAL SETUP

The experimental setup involves assembling all hardware components and testing the system under real operating conditions. The ESP32 microcontroller is programmed to receive commands from the Blynk mobile application.

During operation, the user controls the boat through the mobile interface while observing the live video feed from the ESP32-CAM module. As the boat moves across the water surface, floating debris is captured by the net placed at the front of the boat. This setup allows testing of the boat's movement, control accuracy, waste collection efficiency, and real-time monitoring capabilities



4.3.1 Physical Setup Overview

The physical setup includes the complete arrangement of mechanical and electronic components mounted on the floating boat platform. The ESP32 microcontroller, motor driver module, battery, and other electronic components are placed on a protected board to avoid exposure to water.

The DC motors are connected to propellers that generate thrust when powered, allowing the boat to move forward or backward. The waste collection net is fixed at the front of the boat so that floating waste can be easily captured as the boat moves. The ESP32-CAM module is mounted in a position that provides a clear view of the surrounding environment.



4.3.2 Data Flow and Control Process

The system operates based on a simple communication and control flow. When the boat is powered on, the ESP32 microcontroller connects to the Wi-Fi network and establishes communication with the Blynk mobile application.

The user sends movement commands through the mobile interface. These commands are received by the ESP32 and converted into control signals for the motor driver module. The motor driver then activates the DC motors, enabling the boat to move in the desired direction. At the same time, the ESP32-CAM module continuously captures images and streams video to the operator, allowing real-time monitoring of the cleaning process.

Data Flow of the IoT-Based Seashore Waste Collector Boat

1. System Initialization:

- When the system is powered on, the ESP32 microcontroller and other electronic components are activated.
- The ESP32 initializes the motor driver, servo motor, and ESP32-CAM module for operation.

2. Wireless Communication Establishment:

- The ESP32 connects to the Wi-Fi network and establishes communication with the mobile or laptop interface.
- The operator can send control commands to the boat through the IoT application.

3. Live Video Monitoring:

- The ESP32-CAM captures real-time images from the surrounding water surface.
- The captured frames are transmitted as a live video stream through Wi-Fi to the user's device.

4. Navigation Control:

- The operator observes the live video feed and sends navigation commands through the application.
- These commands are received by the ESP32 microcontroller.

5. Motor Driver Activation:

- The ESP32 sends control signals to the L298N motor driver module.
- The motor driver regulates the power supplied to the DC motors.

6. Boat Movement:

- The DC motors rotate the propellers, allowing the boat to move forward or backward on the water surface.
- This movement helps the boat reach areas where floating waste is present.

7. Waste Collection Mechanism:

- As the boat moves forward, floating waste enters the collection net placed at the front of the boat.
- The net traps plastic debris and other floating materials from the water surface.

8. Net Monitoring and Return Operation:

- The operator monitors the filling level of the net using the live video feed.
- Once the net is filled with waste, the operator remotely navigates the boat back to the shore for waste removal.

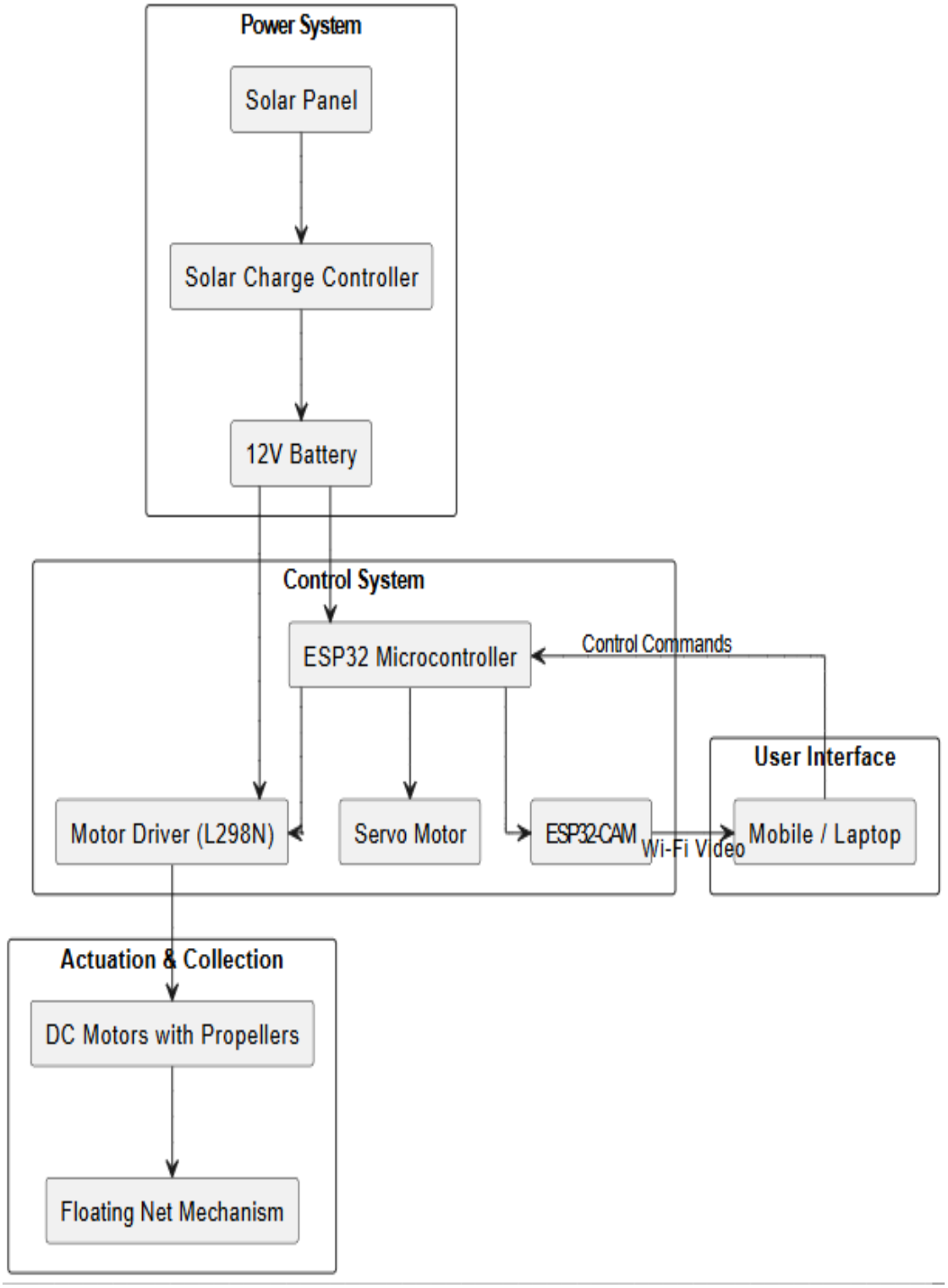
9. Power Management:

- The solar panel generates electrical energy from sunlight.
- The solar charge controller regulates the power and charges the 12V lead-acid battery, which supplies energy to the entire system.

4.4 BLOCK DIAGRAM OF THE SYSTEM

The block diagram of the proposed system illustrates the interaction between the major components. The solar panel and battery provide power to the system through the solar charge controller. The ESP32 microcontroller acts as the central processing unit that receives commands from the Blynk IoT application.

The motor driver module controls the DC motors that drive the propellers, enabling boat movement. The ESP32-CAM module provides live video streaming to the user interface. The floating net mechanism collects waste during the movement of the boat. This block diagram represents the complete functional structure of the waste collector boat.

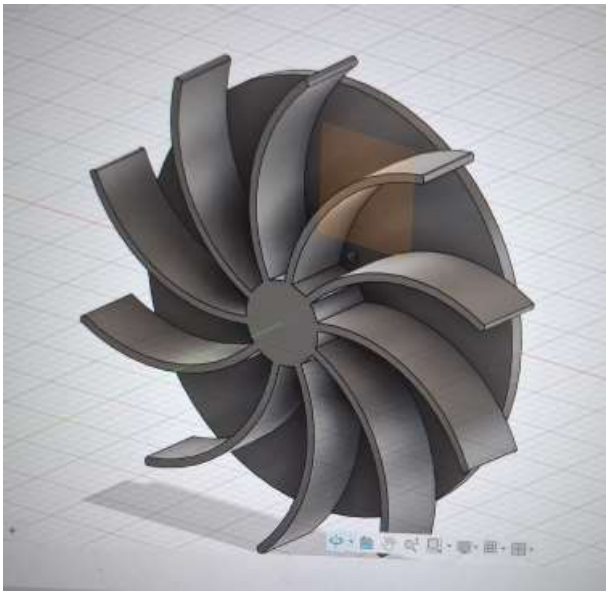

Power System
Solar Panel
Solar Charge Controller
12V Battery
Control System
ESP32 Microcontroller
Control Commands
User Interface
Motor Driver (L298N)
Servo Motor
ESP32-CAM
Wi-Fi Video
Mobile / Laptop
Actuation & Collection
DC Motors with Propellers
Floating Net Mechanism

4.5 MECHANICAL DESIGN

The mechanical design of the boat focuses on maintaining stability, buoyancy, and efficient waste collection. The boat structure is designed to float properly while supporting the weight of electronic components and the collected waste.



The propulsion system is positioned in such a way that the boat can move smoothly across the water surface. The waste collection net is designed to allow water to pass through while trapping floating debris. The design ensures that the boat can operate effectively without losing balance during movement.



4.6 REAL-TIME TESTING

After assembling the system, several tests were conducted to evaluate the performance of the waste collector boat. The tests focused on verifying the functionality of the propulsion system, wireless communication, video monitoring, and waste collection mechanism. During testing, the boat was operated on a water surface while being controlled through the Blynk mobile application. The system successfully responded to user commands and collected floating debris through the net mechanism. The testing results confirmed that the system can effectively assist in removing waste from water surfaces.

4.6.1 Parameters Measured

During the real-time experiments, the following key parameters were observed:

- **Motor Response Time:**

Time taken by the ESP32 to process the command from the Blynk application and activate the motor driver.

- **Wireless Communication Delay:**

Time required for the control command to travel from the mobile application to the ESP32 microcontroller through Wi-Fi.

- **Video Streaming Performance:**

Quality and continuity of the live video stream transmitted from the ESP32-CAM module.

- **Waste Collection Efficiency:**

Ability of the floating net mechanism to capture and retain floating debris during boat movement.

- **System Stability:**

Overall stability of the system during continuous operation, including power supply performance and communication reliability.

4.6.2 Evaluation in Prototype Setup

The system was initially tested in a controlled environment using a prototype water setup. Small floating materials such as plastic pieces and paper fragments were placed on the water surface to simulate real waste conditions. The operator controlled the boat through the Blynk mobile application while monitoring the live video feed from the ESP32-CAM module.

The boat successfully navigated through the water surface and collected floating debris using the front-mounted net mechanism. The ESP32 microcontroller responded quickly to the control signals, and the motor driver operated the DC motors smoothly. The solar-powered battery system provided sufficient power for continuous operation. Minor adjustments were made to the net position and motor alignment to improve stability and collection efficiency.

4.6.3 Real-World Testing Results

In real testing conditions, the waste collector boat demonstrated effective performance in collecting floating waste from the water surface. The live video streaming feature allowed the operator to visually guide the boat toward waste-concentrated areas. The system responded well to control commands and maintained stable communication through the Wi-Fi network.

The floating net mechanism successfully captured lightweight waste materials while the boat moved forward. Once the net was filled, the operator navigated the boat back to the shore for waste removal. The tests confirmed that the system can serve as a practical solution for cleaning small water bodies such as ponds, lakes, and coastal areas.

4.6.4 Comparative Analysis

Compared to traditional manual cleaning methods, the proposed system offers several advantages. Manual waste collection in water bodies is time-consuming, labor-intensive, and sometimes unsafe for workers. The developed waste collector boat reduces human involvement by enabling remote monitoring and control through an IoT platform.

The integration of ESP32, motor driver, solar power system, and camera module provides a compact and efficient solution for environmental maintenance. The system improves waste collection efficiency, enhances monitoring capability, and demonstrates the potential of combining IoT technology with environmental cleaning applications.

4.7 ESP32 PROGRAMMING

The ESP32 microcontroller is programmed to manage all system operations. The program initializes Wi-Fi connectivity, establishes communication with the Blynk IoT platform, and processes commands received from the mobile application.

The microcontroller also controls the motor driver module to regulate the movement of the boat. In addition, it manages the servo motor for camera rotation and coordinates the operation of the ESP32-CAM module for video streaming. This programming structure ensures smooth and reliable operation of the entire system.

4.7.1 Updated Logic Flow Using ESP32

- System power supply is switched ON.
- ESP32 microcontroller initializes all connected modules.
- The ESP32 connects to the Wi-Fi network and establishes communication with the Blynk IoT application.
- The user sends control commands through the mobile application.
- ESP32 receives the commands and sends control signals to the motor driver module.
- The motor driver activates the DC motors to move the boat in the desired direction.
- The ESP32-CAM module captures and streams live video to the user interface.

- Floating waste is collected through the net mechanism during forward motion.
- The operator monitors the waste collection through live video.
- When the net becomes full, the boat is remotely navigated back to the shore.

4.7.2 ESP32 Coding Interface with Blynk IOT

```
/************ BLYNK DETAILS ************/
#define BLYNK_TEMPLATE_ID "TMPL3JutiLaqv"
#define BLYNK_TEMPLATE_NAME "seashore"
#define BLYNK_AUTH_TOKEN "ukiA4rH24FPP_R7KVn0E1cikyPOV6e7p"

/************ LIBRARIES ************/
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
#include <ESP32Servo.h>

/************ WIFI ************/
char ssid[] = "mskkan143";
char pass[] = "123456789";

/************ L298N MOTOR PINS (NO EN PIN) ************/
#define IN3 26
#define IN4 27

/************ SERVO ************/
#define SERVO_PIN 13
Servo myServo;

/************ SETUP ************/
void setup() {
  Serial.begin(115200);

  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);

  // Motor STOP at boot
  digitalWrite(IN3, LOW);
```

```
  digitalWrite(IN4, LOW);

  // Servo setup
  myServo.setPeriodHertz(50);
  myServo.attach(SERVO_PIN, 500, 2400);
  myServo.write(0);   // Initial position

  WiFi.begin(ssid, pass);
  Serial.print("Connecting to WiFi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\n✅ WiFi Connected");

  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
}
/************ BLYNK CONTROLS ************/
// 🔼 FORWARD (V0)
BLYNK_WRITE(V0) {
  int value = param.asInt();
  if (value == 1) {
    digitalWrite(IN3, HIGH);
    digitalWrite(IN4, LOW);
    Serial.println("Motor Forward");
  } else {
    // Stop when button released
    digitalWrite(IN3, LOW);
    digitalWrite(IN4, LOW);
  }
}
// 🔽 REVERSE (V1)
BLYNK_WRITE(V1) {
  int value = param.asInt();
  if (value == 1) {
    digitalWrite(IN3, LOW);
    digitalWrite(IN4, HIGH);
```

```
    Serial.println("Motor Reverse");
  } else {
    // Stop when button released
    digitalWrite(IN3, LOW);
    digitalWrite(IN4, LOW);
  }
}
// 🔄 SERVO CONTROL (V3)
BLYNK_WRITE(V3) {
  int value = param.asInt();
  if (value == 1) {
    myServo.write(90);   // Open
    Serial.println("Servo Open");
  } else {
    myServo.write(0);    // Close
    Serial.println("Servo Close");
  }
}
/************ LOOP ************/
void loop() {
  Blynk.run();
}
```

4.7.3 ESP32 CAM Coding

```
#include "esp_camera.h"
#include <WiFi.h>
#include "esp_http_server.h"

/* ===== Wi-Fi Credentials ===== */
const char* ssid = "mskkan143";         // replace with your Wi-Fi or hotspot
const char* password = "123456789";

/* ===== AI THINKER ESP32-CAM Pins ===== */
#define PWDN_GPIO_NUM     32
#define RESET_GPIO_NUM    -1
#define XCLK_GPIO_NUM      0
#define SIOD_GPIO_NUM     26
#define SIOC_GPIO_NUM     27
```

```
#define Y9_GPIO_NUM       35
#define Y8_GPIO_NUM       34
#define Y7_GPIO_NUM       39
#define Y6_GPIO_NUM       36
#define Y5_GPIO_NUM       21
#define Y4_GPIO_NUM       19
#define Y3_GPIO_NUM       18
#define Y2_GPIO_NUM        5
#define VSYNC_GPIO_NUM    25
#define HREF_GPIO_NUM     23
#define PCLK_GPIO_NUM     22

httpd_handle_t stream_httpd = NULL;

/* ===== STREAM HANDLER ===== */
static esp_err_t stream_handler(httpd_req_t *req) {
  camera_fb_t * fb = NULL;
  char part_buf[64];

  httpd_resp_set_type(req, "multipart/x-mixed-replace;boundary=frame");

  while (true) {
    fb = esp_camera_fb_get();
    if (!fb) {
      Serial.println("❌ Camera capture failed → Restarting");
      ESP.restart();
    }

    size_t hlen = snprintf(part_buf, 64,
      "--frame\r\nContent-Type: image/jpeg\r\nContent-Length: %u\r\n\r\n",
      fb->len);

    httpd_resp_send_chunk(req, part_buf, hlen);
    httpd_resp_send_chunk(req, (const char *)fb->buf, fb->len);
    httpd_resp_send_chunk(req, "\r\n", 2);

    esp_camera_fb_return(fb);
    delay(80);  // safe for hotspot
  }
}

/* ===== START CAMERA SERVER ===== */
void startCameraServer() {
  if (stream_httpd != NULL) return; // already running
```

```
  httpd_config_t config = HTTPD_DEFAULT_CONFIG();
  config.server_port = 81;

  httpd_uri_t stream_uri = {
    .uri      = "/stream",
    .method   = HTTP_GET,
    .handler  = stream_handler,
    .user_ctx = NULL
  };

  if (httpd_start(&stream_httpd, &config) == ESP_OK) {
    httpd_register_uri_handler(stream_httpd, &stream_uri);
    Serial.println("✅ Camera server started at /stream");
  }
}

void setup() {
  Serial.begin(115200);
  Serial.println();

  camera_config_t config;
  config.ledc_channel = LEDC_CHANNEL_0;
  config.ledc_timer   = LEDC_TIMER_0;
  config.pin_d0       = Y2_GPIO_NUM;
  config.pin_d1       = Y3_GPIO_NUM;
  config.pin_d2       = Y4_GPIO_NUM;
  config.pin_d3       = Y5_GPIO_NUM;
  config.pin_d4       = Y6_GPIO_NUM;
  config.pin_d5       = Y7_GPIO_NUM;
  config.pin_d6       = Y8_GPIO_NUM;
  config.pin_d7       = Y9_GPIO_NUM;
  config.pin_xclk     = XCLK_GPIO_NUM;
  config.pin_pclk     = PCLK_GPIO_NUM;
  config.pin_vsync    = VSYNC_GPIO_NUM;
  config.pin_href     = HREF_GPIO_NUM;
  config.pin_sccb_sda = SIOD_GPIO_NUM;
  config.pin_sccb_scl = SIOC_GPIO_NUM;
  config.pin_pwdn     = PWDN_GPIO_NUM;
  config.pin_reset    = RESET_GPIO_NUM;

  config.xclk_freq_hz = 10000000;
  config.pixel_format = PIXFORMAT_JPEG;
  config.frame_size   = FRAMESIZE_QVGA;  // safe for hotspot
```

```
  config.jpeg_quality = 15;
  config.fb_count     = 1;

  if (esp_camera_init(&config) != ESP_OK) {
    Serial.println("❌ Camera init failed → Restarting");
    ESP.restart();
  }

  WiFi.begin(ssid, password);
  Serial.print("📶 Connecting to Wi-Fi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

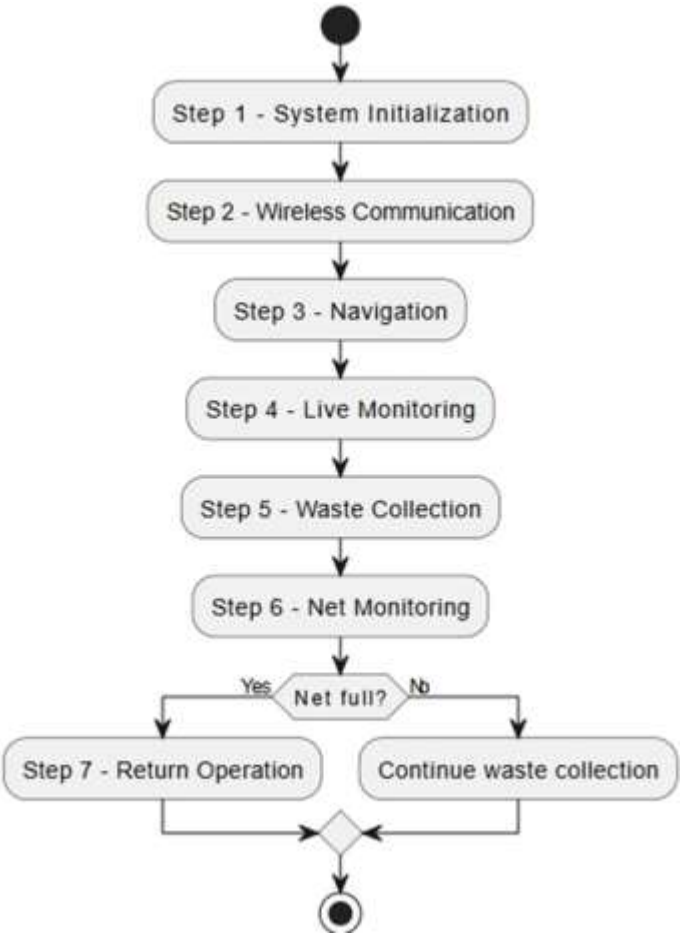
  Serial.println("\n✅ Wi-Fi connected!");
  Serial.print("🌐 ESP32-CAM IP: ");
  Serial.println(WiFi.localIP());

  Serial.print("📺 Live Stream URL:\nhttp://");
  Serial.print(WiFi.localIP());
  Serial.println(":81/stream");

  startCameraServer();
}

void loop() {
  delay(1);
}
```

4.7.4 Flowchart for Algorithm



4.8 SUMMARY

This chapter discussed the implementation framework of the proposed IoT-Based Seashore Waste Collector Boat. The system architecture, hardware components, software structure, and experimental setup were explained in detail. The chapter also described the mechanical design, system testing process, and programming approach used for the ESP32 microcontroller. The implementation framework demonstrates how the integration of IoT technology, embedded systems, and renewable energy can be used to develop an efficient waste collection system for water bodies. The next chapter focuses on the preparation and integration of mechatronic components used in the project.

CHAPTER – 5

MECHATRONIC COMPONENT PREPARATION

5.1 3D PRINTED PROPELLER FABRICATION

The propulsion system of the waste collector boat requires an efficient propeller to generate thrust and enable movement on the water surface. In this project, the propeller was designed using 3D modeling software and fabricated using a 3D printing process. The design was optimized to ensure lightweight structure, balanced rotation, and adequate thrust generation for the DC motors used in the boat.

The 3D printed propeller was manufactured using PLA material, which provides sufficient strength while maintaining low weight. After fabrication, the propeller was attached to the shaft of the DC gear motor. This arrangement allows the motor to rotate the propeller and push water backward, thereby moving the boat forward. The use of 3D printing enabled quick customization of the propeller design and improved the overall efficiency of the propulsion system.



5.2 ESP32-CAM PROTECTIVE ENCLOSURE DESIGN (3D PRINTING)

To protect the camera module from environmental conditions such as water splashes and physical damage, a dedicated enclosure was designed for the ESP32-CAM module. The enclosure was modeled using 3D design software and fabricated using a 3D printing technique.

The enclosure was designed with proper openings for the camera lens, ventilation, and wiring connections. This protective housing ensures stable positioning of the camera while allowing clear video capture. The ESP32-CAM module placed inside the enclosure streams live video to the user interface, enabling real-time monitoring of the waste collection process. The use of a 3D printed enclosure improves durability and enhances the overall reliability of the monitoring system.



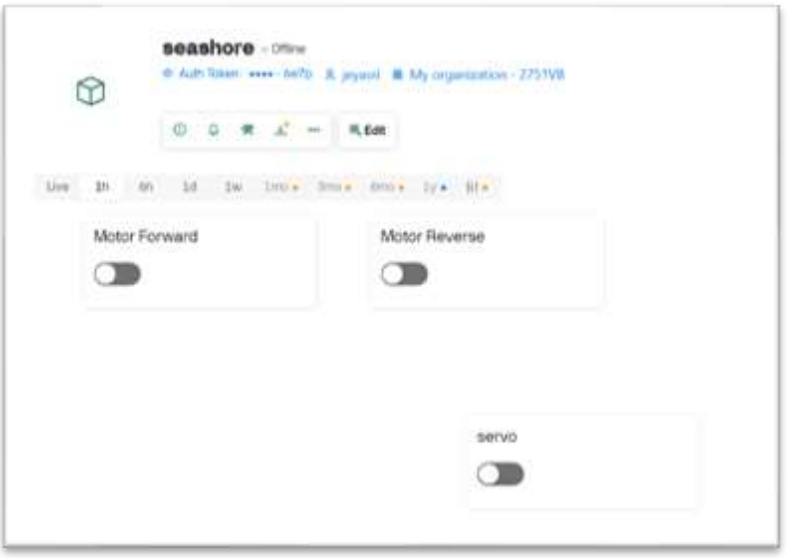
5.3 BLYNK IOT INTEGRATION FOR BOAT CONTROL

The control system of the waste collector boat is implemented using the Blynk IoT platform. Blynk provides a user-friendly mobile interface that allows the operator to remotely control the movement of the boat and monitor system operations.

In this project, the ESP32 microcontroller is connected to the Blynk IoT application through a Wi-Fi network. The mobile application includes control buttons for forward movement, reverse movement, and servo motor operation. When the user presses a control button in the application, the command is transmitted through the internet to the ESP32 microcontroller. The ESP32 processes the command and activates the motor driver or servo motor accordingly.

This IoT-based control system enables wireless operation of the boat and allows the user to monitor and control the waste collection process from a mobile device, making the system more efficient and convenient.



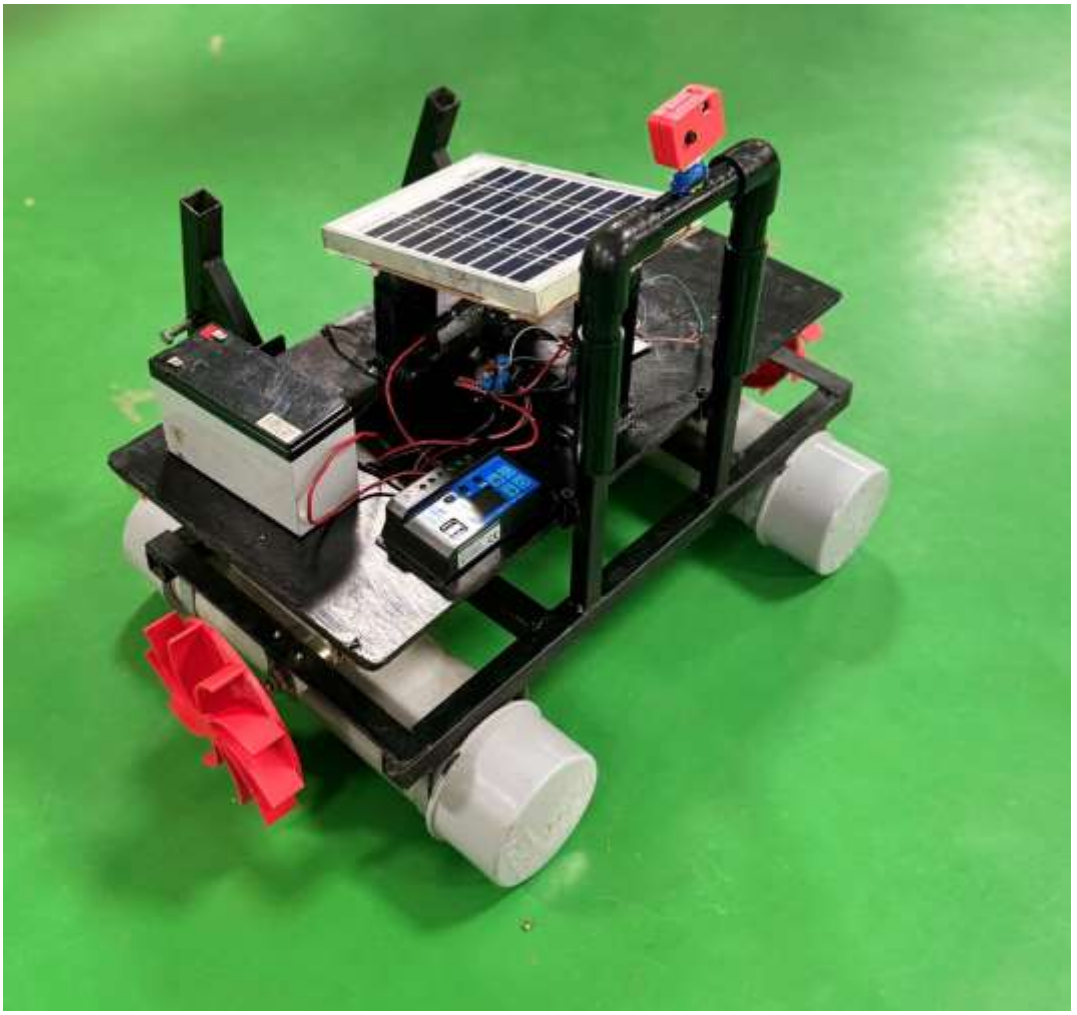




5.3 WASTE COLLECTOR BOAT SETUP

After assembling the system, several tests were conducted to evaluate the performance of the waste collector boat. The tests focused on verifying the functionality of the propulsion system, wireless communication, video monitoring, and waste collection mechanism.

During testing, the boat was operated on a water surface while being controlled through the Blynk mobile application. The system successfully responded to user commands and collected floating debris through the net mechanism. The testing results confirmed that the system can effectively assist in removing waste from water surfaces.



CHAPTER - 6
CONCLUSION AND SCOPE FOR FUTURE WORK

6.1 CONCLUSION

The IoT-based seashore waste collector boat was successfully designed and developed to address the problem of floating waste accumulation in water bodies. The system integrates mechanical, electrical, and IoT technologies to create a practical solution for environmental cleaning. The ESP32 microcontroller acts as the central control unit, enabling communication between the boat and the mobile application through the Blynk IoT platform. Using this interface, the operator can remotely control the movement of the boat and monitor the waste collection process.

The integration of the ESP32-CAM module provides real-time video streaming, allowing the user to observe the surrounding water surface and guide the boat effectively. The floating net mechanism collects plastic waste and other floating debris during the boat's movement. The use of a solar charging system with a lead-acid battery ensures a sustainable and energy-efficient power supply for the system. Experimental testing demonstrated that the prototype can successfully navigate the water surface and collect lightweight waste materials. The developed system reduces manual effort and provides an efficient approach for maintaining cleaner water bodies.

6.2 SCOPE FOR FUTURE WORK

Although the current prototype demonstrates the feasibility of the proposed system, several improvements can be implemented in the future to enhance performance, autonomy, and scalability.

6.2.1 Integration with Advanced Vision Systems

Future versions of the system can incorporate advanced computer vision techniques to automatically detect floating waste. By using image processing or machine learning models, the camera module could identify waste materials and guide the boat toward polluted areas. This would reduce the need for manual navigation and improve cleaning efficiency.

6.2.2 Cloud-Based Monitoring and Data Management

The system can be enhanced by integrating cloud-based monitoring platforms. Data such as waste collection statistics, boat location, and operational status can be uploaded to cloud servers for remote monitoring and analysis. This would allow environmental agencies and researchers to track pollution levels and evaluate the effectiveness of cleaning operations.

6.2.3 Autonomous Navigation and Path Planning

The current system relies on manual control through a mobile application. Future improvements could include autonomous navigation using GPS modules and obstacle detection sensors. Implementing intelligent path planning algorithms would enable the boat to automatically cover large water areas and optimize the waste collection process.

6.3 FINAL REMARKS

The proposed IoT-based seashore waste collector boat demonstrates the potential of combining robotics, IoT technology, and renewable energy to address environmental challenges. The system offers a low-cost and efficient solution for cleaning floating waste in small water bodies such as ponds, lakes, and coastal areas. With further technological enhancements and large-scale deployment, such systems can contribute significantly to maintaining cleaner aquatic environments and promoting sustainable waste management practices.

6.5 REFERENCES

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ANNEXURE – I

Appendices

• Arduino (ESP32) program code used for controlling the waste collector boat

• Photo of the complete prototype of the IoT-based seashore waste collector boat

• Internal view showing electronic components such as ESP32, motor driver, battery, and wiring connections

• Images of 3D printed components (propeller and ESP32-CAM protective enclosure)

• Installation of the floating waste collection net mechanism

• Screenshot of the Blynk IoT application interface used for boat control

• List of hardware components used in the system

• Wiring diagram of ESP32, motor driver, servo motor, and ESP32-CAM module

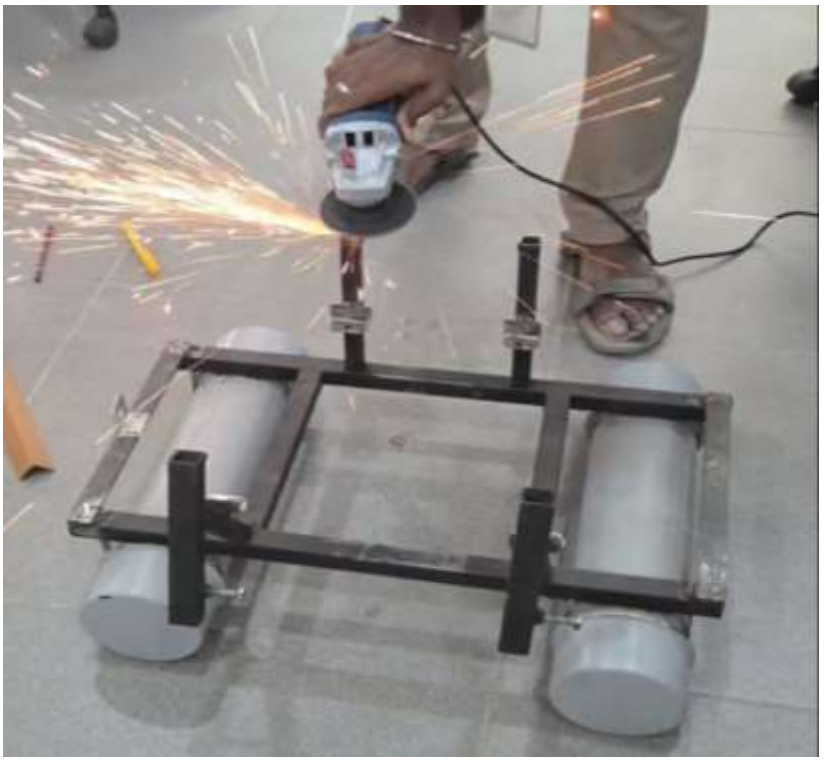
• Solar panel and charge controller connection diagram

• Step-by-step assembly process of the boat prototype with photographs

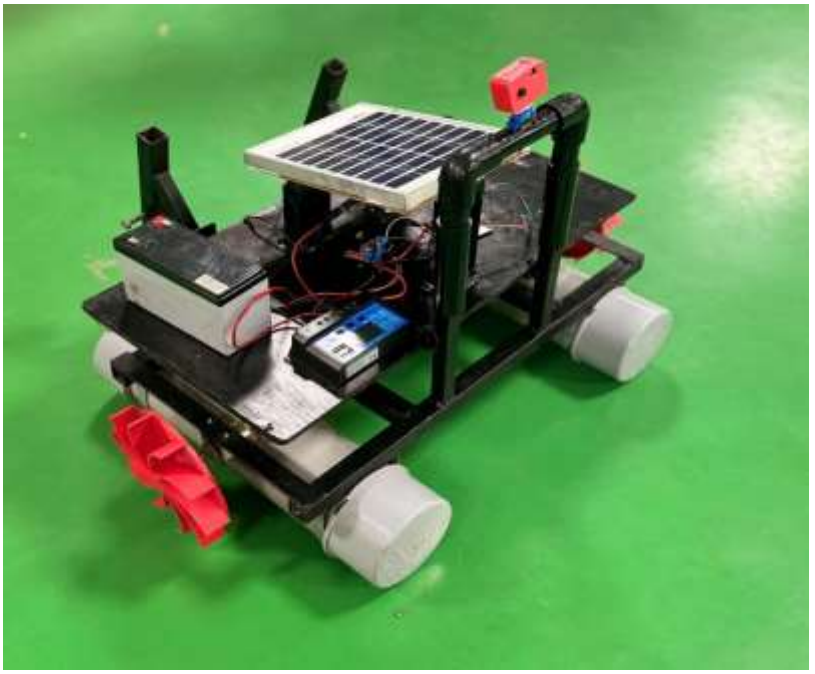
BOAT STRUCTURE



STRUCTURE FABRICATION



BOAT FULL PROTOTYPE



ANNEXURE - II

BASE PAPER

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Remote Controlled Unmanned River Cleaning Bot

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Abstract- **This project focuses more on "Remote Controlled Unmanned River Cleaning Bot". In India water pollution is increasing day by day so this is often becoming a significant issue for rivers, ponds etc. This mainly consists of impurities like wastewater debris, plastics, garbage on floating water surfaces. These impurities mainly affect the health of peopleand also affect the life of aquatic animals. Normally this project is based on renewable energy sources, so there's reduction in use of non renewable energy sources like oil, petroleum, electricity and all types of mineral sources. So by this non renewable energy sources are saved. So this project helps to reduce the water pollution on floating bodies.**

Keywords - Water debris, aquatic animals, pollution, garbage

I. INTRODUCTION

Generally, conventional method based on manual basis and it's used for collection of water debris, trash, plastic and all other sorts of impurities which is floating on water bodies or by collecting this impurity by means of boat, thrash skimmer etc. And removed this impurity near the river shore and disposed of it. But this conventional method requires more manpower; hence this is often a risky, costly and time consuming method. By considering this all remote operated floating river cleaning machines are more efficient than conventional methods and also this is effective and eco-friendly.



Fig 1 : Water Pollution

This machine is remote operated so manpower isn't required at all. So this machine is really advantageous for reducing the pollution on Ganga river which is caused by 'Kumbh Mela' And also Government of India has taken charge to wash river and pond thanks to increasing water pollution , then that they invest huge capital for several river cleaning project like 'Namami Ganga', 'Narmada bachao'. And also developed many projects in various cities like Ahmedabad, Varanasi etc. By taking into consideration, this Remote operated river cleaning machine has been designed to clean river floating surfaces.

By taking into consideration, this Remote Controlled unmannedriver cleaning machine has been designed to clean river floatingsurfaces.

II. MOTIVATION

As India is developing in terms of technology very rapidly,it is quite important that the devices which are introduced need to be updated on a regular basis. Nowadays, all the devices are more dependent on digital technology rather than analog technology. So, people in the government cleaning sector can start using this bot in their Cleaning activities.

III. PROBLEM DEFINITION

In the absence of garbage disposal facilities, the practice of dumping garbage into nearby water bodies has become quite common in recent years and has posed long-term negative impacts both on biodiversity of the area and as well as on the local environment.

IV. SYSTEM DESCRIPTION

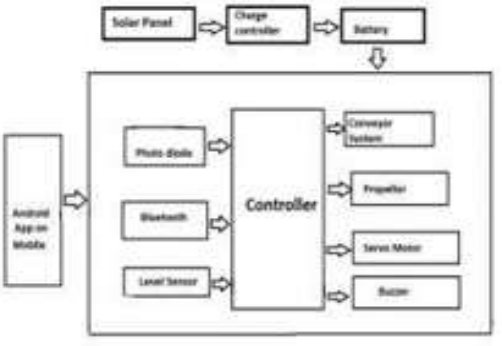


Fig 2 : Block Diagram

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We know that solar panels convert light energy into electrical energy which is DC in nature. Solar panel output energy is given to the charge controller that controls DC output of solarwhich is pulsating in nature and fed pure DC to charge the battery. As we know that battery is used to store the energy. This stored energy is given to all this circuitry for overall operation. This circuitry consists of a controller, level sensor, photodiodes, Bluetooth, conveyor system, propeller, buzzer, and servo motor. In this system controller is a main part it's having input like photodiode, Bluetooth, level sensor and output consist of conveyor system, propeller, buzzer, and servo motor. Android app is used for overall remote controlling purposes.

V. WORKING

In this project the foremost aim of this machine is to lift waste debris from the water surface and dispose of it within the tray. It consists of an arrangement of conveyor which is placed on the shaft of the motor. Due rotation of motor conveyor rotated. Because the conveyor is moved, it collects water debris, waste garbage and plastics from water bodies. because the machine is placed within the water the waste debris in water will get lifted and it moves in an upward direction. because the waste debris reaches the upper extreme position it'll get dropped within the tray. Hence this will end in cleaning of water surfaces and safe collection of waste debris from water. Propeller is used to drive the machine on the river and run with the help of a PMDC motor. The total electrical devices are controlled by an RF transmitter and receiver which are used to manage the machine remotely.

Collecting Mechanism is employed in our project to beat real time issues as thanks to water tension garbage is difficult to collect. By using this four bar mechanism, it rotated at a particular angle intended to gather the rubbish for the model. it's two windows open and shut as the user wishes using remote to ON and OFF the mechanism. Water wheel is bolted on a shaft which is placed aboard the frame. The aim of a water wheel (propeller) is to maneuver the machine forward or backward on water. Motor is used to rotate the water wheel with the assistance of a chain drive mechanism. In this project tracking system is additionally implemented which is helpfulto regulate angle of solar array with reference to sunrays. So that we get more solar output.

VI. COMPONENTS USED

1) Arduino Board: Arduino Uno board is based on the ATmega328P. It consists of 14 digital input/output pins in which6 can be used as PWM outputs, 6 analog inputs, 16 MHZ quartzcrystal, a USB connection, a power jack, an ICSP header and a reset button.

2) L293D Motor Driver Board: The 293D is used to provide bidirectional drive current up to 600mA and voltage from 5V to 36V.L293D consist of the output clamping diodes for protection.

3) HC 05 Bluetooth module: This Bluetooth module is designed for transparent wireless serial connection. This can be used in a Master or Slave configuration; it makes a good solution for wireless communication.

4) DC motor: In this machine two PMDC motors are used and these motors are used to control the direction of the propeller. Another BO (battery operated) DC motor used to control the conveyor belt and also a DC servomotor used for tracking systems.

5) Battery: This is a lead acid rechargeable battery. This machine consists of four batteries which gives output such as 2A, 8V for operation of setup.

6) Relay Module: This module provides the protection to the microcontroller from the higher load current.

7) Solar Panel: A solar panel consists of photovoltaic cells, which can be used to generate electricity through photovoltaic effect. This energy used to charge the batteries. Solar output is given to the DC Regulators.

8) Conveyor Belt: In this machine we used the polyvinyl Conveyor Belt. This is controlled by the arduino uno system using a motor driver circuit. This collects all floating waste from the water surface and discharges it into the dustbin.

9) Blucontrol [android application]: This android application is installed in mobile phones to control the setup automatically which can be downloaded from the android app market for free of cost.

VII. DESIGN CALCULATIONS

Battery-
8V/2Am
pPanel-
12V/3wa
tt
Current (I) = P/V
I = 3/12I =
0.25Amp
Charging Voltage = 9V/250mA
Charging Time = (Battery Watt/Panel Watt) *2 =
(16/2.25)*2
= 14.22hrs
Discharge Time = (Battery Amp Hr/Total current
Consumed)
= 2000mA/1270
=1.57hrs

VIII. OBJECTIVES

- To reduce man power
- Reduce time consumption.
- Reduce pollution.
- Increase efficiency.
- Cost effectiveness.
- Increase reliability.
- Increase life of aquatic animals.
- Reduce use of non renewable energy sources.

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- Reduce the pollution of Godavari river.
- Maintain a clean and healthy environment.

IX. ADVANTAGES

- This cleaning system is easy to operate and flexible.
- This system is Eco-friendly.
- This requires less manpower.
- This required more use of renewable energy Sources.
- This system is Cost effective (Initial and Maintenance costis low).
- This is efficient method

X. APPLICATIONS

- Useful to reduce the water pollution in river
- It is applicable to reduce water debris, impurities, and all types of impurities which are floating on the water surface in swimming pools.
- It is useful to remove the environmental marine pollutionat Godavari River.
- It is useful in fishery plants to collect dead fishes.

XI. DISADVANTAGES

- Waste collecting capacity is limited.
- Only useful to collect waste which is floating on riversurface

XII. CONCLUSION

This project emphasizes supply flexibility in operation. This is often easy to operate and the price of maintenance is low. Hencethis project "Remote Controlled Unmanned Floating River Cleaning Machine" is usually designed to form a system considerably economical and helpful to get rid of water
impurities like plastics, trashes, water debris which is floating on river and pond surface.This is mainly very useful maintaining human health and for increasing the lifetime of aquatic animals.

XIII. FUTURE SCOPE

Now day by day the world is facing the biggest problem of floating garbage. And it's increasing in tremendous amounts so it's very difficult to wash all this floating garbage due to more requirement of manpower. so, in future this remote operated floating river cleaning machine has more scope to remove large capacity of garbage automatically as fast as possible. And by making modifications during this machine, this is used for automatically removing the garbage from beaches also.

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ANNEXURE – III

PLAGIARISM REPORT